

In [32]: **import** pandas **as** pd

```
# Replace 'Netflix.csv' with the correct file path if it's not in the same directory
df = pd.read_csv('Netflix.csv')

# a) Print the first 5 rows and last 5 rows of the dataframe
print(df.head())
print(df.tail())

# b) Check number of rows and columns
num_rows, num_columns = df.shape
print(f"Rows: {num_rows}, Columns: {num_columns}")

# c) Print all column names
print("Column Names:", df.columns.tolist())

# d) Calculate descriptive statistics of all variables
descriptive_stats = df.describe(include='all')
print(descriptive_stats)

# e) Check the number of unique values for each column
unique_values = df.nunique()
print("Unique values per column:\n", unique_values)

# f) Check the percentage of missing values for each column
missing_percentage = df.isnull().mean() * 100
print("Missing values percentage:\n", missing_percentage)

# g) Delete all rows where Director column has missing values
df_cleaned = df.dropna(subset=['director'])

# h) Print all records where country contains "Germany" (including West Germany)
df_germany = df[df['country'].str.contains('Germany', na=False, case=False)]
print(df_germany)

# i) Expand Duration column into two separate columns (numeric and string)
df[['duration_value', 'duration_unit']] = df['duration'].str.extract(r'(\d+)([a-zA-Z]+)')

# j) Split Date Added column into three separate columns (Day, Month, Year)
df[['month_added', 'day_added', 'year_added']] = df['date_added'].str.split('-', expand=True)

# k) Count the number of TV shows and movies released in each year
release_counts = df['release_year'].value_counts().sort_index()
print("Releases per year:\n", release_counts)

# l) Rename column 'title' to 'movie_title'
df.rename(columns={'title': 'movie_title'}, inplace=True)

# m) Split Listed_in column into three different columns (Genre1, Genre2, Genre3)
df[['Genre1', 'Genre2', 'Genre3']] = df['listed_in'].str.split('|', expand=True)

# Display the first few rows of the modified dataset
print(df.head())
```

	show_id	type	title	director \
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson
1	s2	TV Show	Blood & Water	NaN
2	s3	TV Show	Ganglands	Julien Leclercq
3	s4	TV Show	Jailbirds New Orleans	NaN
4	s5	TV Show	Kota Factory	NaN

	cast	country \
0	NaN	United States
1	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa
2	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN
3	NaN	NaN
4	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India

	date_added	release_year	rating	duration \
0	September 25, 2021	2020	PG-13	90 min
1	September 24, 2021	2021	TV-MA	2 Seasons
2	September 24, 2021	2021	TV-MA	1 Season
3	September 24, 2021	2021	TV-MA	1 Season
4	September 24, 2021	2021	TV-MA	2 Seasons

	listed_in
0	Documentaries
1	International TV Shows, TV Dramas, TV Mysteries
2	Crime TV Shows, International TV Shows, TV Act...
3	Docuseries, Reality TV
4	International TV Shows, Romantic TV Shows, TV ...

	show_id	type	title	director \
8775	s8803	Movie	Zodiac	David Fincher
8776	s8804	TV Show	Zombie Dumb	NaN
8777	s8805	Movie	Zombieland	Ruben Fleischer
8778	s8806	Movie	Zoom	Peter Hewitt
8779	s8807	Movie	Zubaan	Mozez Singh

	cast	country \
8775	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States
8776	NaN	NaN
8777	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States
8778	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States
8779	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India

	date_added	release_year	rating	duration \
8775	November 20, 2019	2007	R	158 min
8776	July 1, 2019	2018	TV-Y7	2 Seasons
8777	November 1, 2019	2009	R	88 min
8778	January 11, 2020	2006	PG	88 min
8779	March 2, 2019	2015	TV-14	111 min

	listed_in
8775	Cult Movies, Dramas, Thrillers
8776	Kids' TV, Korean TV Shows, TV Comedies
8777	Comedies, Horror Movies
8778	Children & Family Movies, Comedies
8779	Dramas, International Movies, Music & Musicals

Rows: 8780, Columns: 11

Column Names: ['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added', 'release_year', 'rating', 'duration', 'listed_in']

	show_id	type	title	director \
count	8780	8780	8780	6150
unique	8780	2	8780	4516

top	s1	Movie	Dick Johnson Is Dead	Rajiv Chilaka
freq	1	6110	1	19
mean	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN

	cast	country	date_added	release_year
\				
count	7956	7952	8770	8780.000000
unique	7671	745	1713	NaN
top	David Attenborough	United States	January 1, 2020	NaN
freq	19	2809	109	NaN
mean	NaN	NaN	NaN	2014.178474
std	NaN	NaN	NaN	8.827938
min	NaN	NaN	NaN	1925.000000
25%	NaN	NaN	NaN	2013.000000
50%	NaN	NaN	NaN	2017.000000
75%	NaN	NaN	NaN	2019.000000
max	NaN	NaN	NaN	2021.000000

	rating	duration	listed_in
count	8776	8780	8780
unique	14	220	514
top	TV-MA	1 Season	Dramas, International Movies
freq	3198	1788	361
mean	NaN	NaN	NaN
std	NaN	NaN	NaN
min	NaN	NaN	NaN
25%	NaN	NaN	NaN
50%	NaN	NaN	NaN
75%	NaN	NaN	NaN
max	NaN	NaN	NaN

Unique values per column:

show_id	8780
type	2
title	8780
director	4516
cast	7671
country	745
date_added	1713
release_year	74
rating	14
duration	220
listed_in	514

dtype: int64

Missing values percentage:

show_id	0.000000
type	0.000000
title	0.000000
director	29.954442
cast	9.384966
country	9.430524
date_added	0.113895
release_year	0.000000
rating	0.045558
duration	0.000000

listed_in 0.000000
dtype: float64

	show_id	type	title \
7	s8	Movie	Sankofa
12	s13	Movie	Je Suis Karl
129	s130	Movie	An Unfinished Life
142	s143	Movie	Freedom Writers
172	s173	Movie	School of Rock
...	...	...	...
8590	s8618	Movie	Trash
8634	s8662	Movie	Unfinished Song
8641	s8669	Movie	V for Vendetta
8702	s8730	Movie	Where the Money Is
8718	s8746	Movie	Willy Wonka & the Chocolate Factory

	director	ast \
7	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...
12	Christian Schwochow	Luna Wedler, Jannis Niewöhner, Milan Peschel, ...
129	Lasse Hallström	Robert Redford, Jennifer Lopez, Morgan Freeman, ...
142	Richard LaGravenese	Hilary Swank, Patrick Dempsey, Scott Glenn, Im...
172	Richard Linklater	Jack Black, Joan Cusack, Mike White, Sarah Silver...
...	...	...
...	...	...
8590	Stephen Daldry	Wagner Moura, Martin Sheen, Rooney Mara, Selto...
8634	Paul Andrew Williams	Terence Stamp, Gemma Arterton, Christopher Ecc...
8641	James McTeigue	Natalie Portman, Hugo Weaving, Stephen Rea, St...
8702	Marek Kaniévská	Paul Newman, Linda Fiorentino, Dermot Mulroney, ...
8718	Mel Stuart	Gene Wilder, Jack Albertson, Peter Onorati, Ro...

	country	date_added
7	United States, Ghana, Burkina Faso, United Kingdom, ...	September 24, 2021
12	Germany, Czech Republic	September 23, 2021
129	Germany, United States	September 1, 2021
142	Germany, United States	September 1, 2021
172	United States, Germany	September 1, 2021
...	...	...
...	...	...
8590	United Kingdom, Brazil, Germany	January 1, 2019
8634	United Kingdom, Germany	July 22, 2019
8641	United States, United Kingdom, Germany	October 1, 2018

87020	Germany, United States, United Kingdom, Canada	January 15, 202
87180	United States, East Germany, West Germany	January 1, 202

	release_year	rating	duration	\
7	1993	TV-MA	125 min	
12	2021	TV-MA	127 min	
129	2005	PG-13	108 min	
142	2007	PG-13	124 min	
172	2003	PG-13	110 min	
...	...	...	...	
8590	2014	R	114 min	
8634	2012	PG-13	94 min	
8641	2005	R	132 min	
8702	2000	PG-13	89 min	
8718	1971	G	100 min	

	listed_in
7	Dramas, Independent Movies, International Movies
12	Dramas, International Movies
129	Dramas
142	Dramas
172	Comedies, Music & Musicals
...	...
8590	Dramas, Independent Movies, Thrillers
8634	Comedies, Dramas, Independent Movies
8641	Action & Adventure, Dramas, Sci-Fi & Fantasy
8702	Action & Adventure, Comedies, Dramas
8718	Children & Family Movies, Classic Movies, Come...

[231 rows x 11 columns]

Releases per year:

release_year	
1925	1
1942	2
1943	3
1944	3
1945	4
...	
2017	1030
2018	1141
2019	1024
2020	952
2021	592

Name: count, Length: 74, dtype: int64

	show_id	type	movie_title	director	\
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	
1	s2	TV Show	Blood & Water	NaN	
2	s3	TV Show	Ganglands	Julien Leclercq	
3	s4	TV Show	Jailbirds New Orleans	NaN	
4	s5	TV Show	Kota Factory	NaN	

	cast	country	\
0	NaN	United States	
1	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	
2	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	
3	NaN	NaN	
4	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	

	date_added	release_year	rating	duration	\
0	September 25, 2021	2020	PG-13	90 min	
1	September 24, 2021	2021	TV-MA	2 Seasons	
2	September 24, 2021	2021	TV-MA	1 Season	
3	September 24, 2021	2021	TV-MA	1 Season	
4	September 24, 2021	2021	TV-MA	2 Seasons	

	listed_in	duration_value	\
0	Documentaries	90	
1	International TV Shows, TV Dramas, TV Mysteries	2	
2	Crime TV Shows, International TV Shows, TV Act...	1	
3	Docuseries, Reality TV	1	
4	International TV Shows, Romantic TV Shows, TV ...	2	

	duration_unit	month_added	day_added	year_added	Genre1
0	min	September	25	2021	Documentaries
1	Seasons	September	24	2021	International TV Shows
2	Season	September	24	2021	Crime TV Shows
3	Season	September	24	2021	Docuseries
4	Seasons	September	24	2021	International TV Shows

	Genre2	Genre3
0	None	None
1	TV Dramas	TV Mysteries
2	International TV Shows	TV Action & Adventure
3	Reality TV	None
4	Romantic TV Shows	TV Comedies

```
In [28]: import pandas as pd
import os

# Load the CSV files
students_df = pd.read_csv('Student.csv')
marks_df = pd.read_csv('marks.csv')

# 1. Combine both dataframes (Full Outer Join)
combined_df = pd.merge(students_df, marks_df, on="Student_id", how="outer")

# 2. Print maximum and minimum marks Gender-wise
gender_marks = combined_df.groupby("Gender")["Mark"].agg(["max", "min"])
print("Maximum and Minimum Marks Gender-wise:\n", gender_marks)

# 3. Students who scored above the class average
average_marks = combined_df["Mark"].mean()
above_avg_students = combined_df[combined_df["Mark"] > average_marks][["S
print("Students who scored above average:\n", above_avg_students)

# 4. Filter Males who are Employed
males_employed = combined_df[(combined_df["Gender"] == "Male") & (combine
print("Males who are Employed:\n", males_employed)

# 5. Create 'IQ_level' Column
def assign_iq_level(marks):
    if marks > 80:
        return "Intelligent"
    elif 50 <= marks <= 80:
        return "Mediocre"
    else:
        return "Weak"
```

```

combined_df["IQ_level"] = combined_df["Mark"].apply(assign_iq_level)

# 6. Count the number of Males and Females from each city
gender_city_count = combined_df.groupby(["City", "Gender"])["Student_id"]
print("Gender count by City:\n", gender_city_count)

# 7. Print the top 5 Male scorers
top_5_male_scorers = combined_df[combined_df["Gender"] == "Male"].nlargest
print("Top 5 Male Scorers:\n", top_5_male_scorers)

# 8. Replace "Male" with "M" and "Female" with "F", then export to CSV
combined_df["Gender"] = combined_df["Gender"].replace({"Male": "M", "Fema

# Ensure the directory exists
directory = 'D:/'
if not os.path.exists(directory):
    os.makedirs(directory)

# Export the DataFrame to CSV
combined_df.to_csv(os.path.join(directory, "test.csv"), index=False)
print("Data exported to D:/test.csv successfully.")

# 9. Check for duplicated Student_IDs
duplicated_students = combined_df[combined_df.duplicated("Student_id", ke
print("Duplicated Student_ids:\n", duplicated_students)

# 10. Create a separate dataframe with only Integer/Float columns
numeric_df = combined_df.select_dtypes(include=["int64", "float64"])
print("Numeric DataFrame:\n", numeric_df.head())

# 11. Student_IDs present in Students table but not in Marks table
students_not_in_marks = students_df[~students_df["Student_id"].isin(marks
print("Student_ids present in Students table but not in Marks table:\n",

```

Maximum and Minimum Marks Gender-wise:

	max	min
Gender		
Female	99.0	40.0
Male	100.0	40.0

Students who scored above average:

	Student_id	Mark
0	1	95.0
2	3	98.0
3	4	75.0
4	5	89.0
11	12	90.0
..	...	...
221	222	74.0
222	223	79.0
224	225	72.0
227	228	99.0
230	231	97.0

[137 rows x 2 columns]

Males who are Employed:

Empty DataFrame

Columns: [Student_id, Age, Gender, Grade, Employed, Mark, City]

Index: []

Gender count by City:

	City	Gender	Student_id
0	Bhopal	Female	13
1	Bhopal	Male	15
2	Chennai	Female	15
3	Chennai	Male	18
4	Delhi	Female	12
5	Delhi	Male	20
6	Gwalior	Female	14
7	Gwalior	Male	18
8	Kochi	Female	10
9	Kochi	Male	21
10	Mumbai	Female	16
11	Mumbai	Male	16
12	Pune	Female	15
13	Pune	Male	26

Top 5 Male Scorers:

	Student_id	Age	Gender	Grade	Employed	Mark	City	IQ_level
149	150	19	Male	1st Class	no	100.0	Bhopal	Intelligent
147	148	18	Male	1st Class	no	99.0	Kochi	Intelligent
2	3	18	Male	1st Class	no	98.0	Mumbai	Intelligent
60	61	18	Male	1st Class	no	98.0	Gwalior	Intelligent
62	63	19	Male	1st Class	no	98.0	Chennai	Intelligent

Data exported to D:/test.csv successfully.

Duplicated Student_ids:

Empty DataFrame

Columns: [Student_id, Age, Gender, Grade, Employed, Mark, City, IQ_level]

Index: []

Numeric DataFrame:

Student_id	Age	Mark
------------	-----	------

0	1	19	95.0
1	2	20	70.0
2	3	18	98.0
3	4	21	75.0
4	5	19	89.0

Student_ids present in Students table but not in Marks table:

8	9
9	10
48	49

Name: Student_id, dtype: int64

In []: Missing values refer to the absence of data **in** one **or** more cells of a dat

Data entry errors
 Incomplete data collection
 Data merging **or** joining mismatches
 Invalid **or** missing responses **in** surveys

we can use several methods to identify missing values **in** a Pandas DataFra

1. Using `isna()` **or** `isnull()`:

Both `isna()` **and** `isnull()` are functions **in** Pandas that **return** a DataFrame python

2. Using `sum()` to Count Missing Values:

3. Using `info()`:

Different Ways of Treating/Imputing/Deleting Missing Values:

Handling missing values can be done **in** several ways depending on the cont

1. Imputation (Filling Missing Values):

Imputation involves filling **in** the missing values **with** some estimated val

Filling **with** a constant value (e.g., 0, 'Unknown', etc.)

Filling **with** the mean, median, **or** mode (**for** numerical data)

Forward fill / Backward fill (filling missing values using previous/next

Example: *# Fill missing values with a constant (e.g., 'Unknown' for st*
`df['City'] = df['City'].fillna('Unknown')`

Fill missing values with the mean (for numeric columns)

`df['Age'] = df['Age'].fillna(df['Age'].mean())`

Forward fill missing values in the 'Name' column

`df['Name'] = df['Name'].fillna(method='ffill')`

Print the updated DataFrame

`print(df)`

2. Deletion (Removing Rows **or** Columns **with** Missing Values):

Another approach **is** to remove rows **or** columns **with** missing values. This **i**

Methods to Delete Missing Data:

Drop rows **with** missing values using `dropna()`.

Drop columns **with** missing values using `dropna(axis=1)`.

example: *# Drop rows with any missing values*

`df_cleaned = df.dropna()`

Drop columns with any missing values

`df_cleaned_columns = df.dropna(axis=1)`

`print("After dropping rows with missing values:\n", df_cleaned)`

`print("After dropping columns with missing values:\n", df_cleaned_columns)`

3. Replacing **with** Statistical Measures (Mean, Median, Mode): For numerical
Example: *# Replace missing values in 'Age' with the median of 'Age'*
`df['Age'] = df['Age'].fillna(df['Age'].median())`

Replace missing values in 'City' with the most frequent value (mode)
`df['City'] = df['City'].fillna(df['City'].mode()[0])`

`print(df)`

4. Forward **and** Backward Filling:

Forward Fill (ffill): Propagates the last valid value forward to fill the

Backward Fill (bfill): Propagates the next valid value backward to fill t

`df['Age'] = df['Age'].fillna(method='ffill')`

Backward fill missing values

`df['City'] = df['City'].fillna(method='bfill')`

`print(df)`